

The effect of job loss on health: evidence from Utah

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Some health and income puzzles

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- ▶ Losing your job causes a 50–100% short-term increase in mortality and a 10–15% long-term increase (Sullivan and von Wachter, 2009)

...which is itself at odds with literature on effect of recessions on health

- ▶ “[A]n increase in the unemployment rate of the magnitude of the Great Recession reduces the average annual age-adjusted mortality rate by 2.3%” (Finkelstein et al., 2025)

What we do

Provide fresh evidence on the effect of employment and income on health in the US

We do so using a mass layoff design and a linkage between unusually rich administrative health data and matched employer-employee data from Utah

- ▶ as seen in [Hong, Holmes, Lavetti, Logan \(2025\)](#) and [Handel, Holmes, Kolstad, Lavetti \(2026\)](#)

We first scrutinize the validity of the mass layoff design

- ▶ Reverse causality bias?
- ▶ Do employers selectively lay off sick workers during mass layoffs?

We then estimate the effect of job loss on mortality, hospitalization, reasons for hospitalization, who pays for hospital care, insurance coverage, consumption of health care services, prescriptions, chronic conditions, health risk, and more!

What we find

Some evidence of reverse causality

- ▶ Even in a mass layoff, displacement selects on workers who coincidentally have a negative health shock before separation
- ▶ However, this is easy to solve: (1) match on prior health; (2) use intent-to-treat

Firms do not appear to selectively lay off sick workers in mass layoffs

- ▶ Separation-health gradient is similar at mass-layoff and control firms

Job loss has large, persistent effects on insurance status and health care

- ▶ Large decline in insurance, driven by loss of employer-sponsored insurance, partially offset by substitution to dependent, Medicaid, and group plans
- ▶ Conditional on showing up in the hospital, workers are much more likely to pay out of pocket
- ▶ Large reduction in doctor visits, screening, and preventive care

...but modest/null effects on health

- ▶ No effect on number of ER/inpatient hospitalizations or mortality
- ▶ No “effect” on number of chronic conditions, health risk, etc.

Literature

Many papers have estimated the effects of job loss on health around the world

- ▶ Europe: Eliason and Storrie (2009), Kuhn, Lalive and Zweimuller (2009), Browning and Heinesen (2012); South Korea: Hwang, Kim and Lee (2025); Brazil: Amorim et al. (2026)
- ▶ Evidence is “mixed”; some papers find modest/null effects

But in the US, health and labor markets are uniquely enmeshed due to prominence of employer-sponsored insurance

Unfortunately, due to data limitations, there is less evidence from the US

- ▶ Sullivan and Von Wachter (2009): admin data + mass layoff design; only effect on mortality; setting = 1980s Pennsylvania
- ▶ Schaller and Stevens (2014): survey data (MEPS, 2-year panel)
- ▶ Michaud, Crimmins, Hurd (2016): survey data (HRS, repeat cross-section)

Our contribution: new evidence from US using rich admin data and carefully scrutinized mass layoff design

Data

Matched employer-employee data (2002–2019)

- ▶ Quarterly earnings, dominant employer ID, worker ID
- ▶ All jobs covered by unemployment insurance in Utah

Census of hospital admissions in Utah (2002–2013)

- ▶ Date of admission, setting (ER, inpatient, ambulatory), diagnoses (ICD), charges, payer

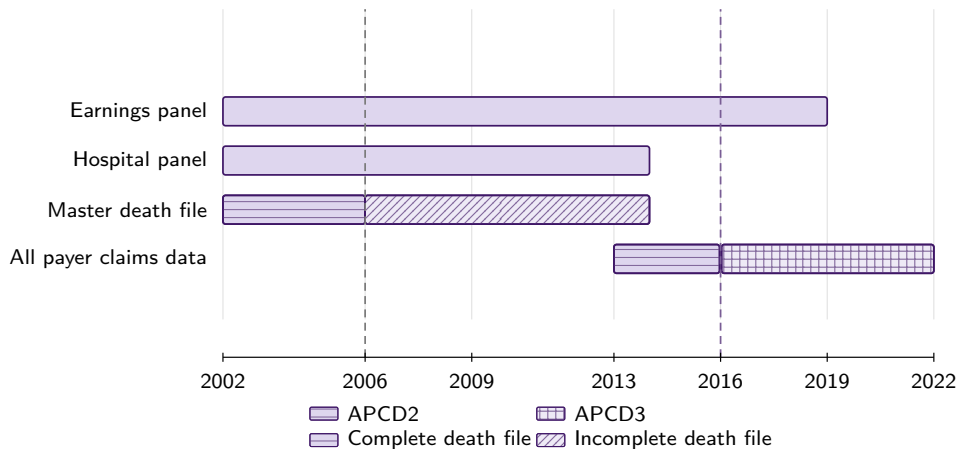
SSA death master file (2002–2013)

- ▶ Universe of deaths until 2006. After 2006, incomplete coverage. We augment with deaths recorded in hospital data.

All payer claims data (2013–2019)

- ▶ Enrollment and claims for all privately insured individuals
- ▶ Measure types of care received, spending, number of chronic conditions, health risk, and more
- ▶ Covers all employers 2013–2015. In 2016, many large self-insuring firms opt out

Data coverage



Mass layoff research design

A **mass layoff** is when employment at a firm with 50 or more workers declines by 30% or more in one year

A worker is **displaced** if they separate from a firm during a mass layoff with at least 3 years of tenure

For every displaced worker, we find one control worker by caliper matching on earnings, firm size, tenure, experience, age, sex, health risk, and hospital visits prior to job loss

We also consider an **intent-to-treat** version, where we compare all incumbent workers at mass-layoff firms—whether or not they separate—with matched control workers. In the intent-to-treat, we do not match on health prior to job loss.

Reverse causality bias

Negative health shocks can cause job loss.

If we misclassify health shocks that coincidentally occur during a mass layoff as effects of job loss, we get **reverse causality bias**.

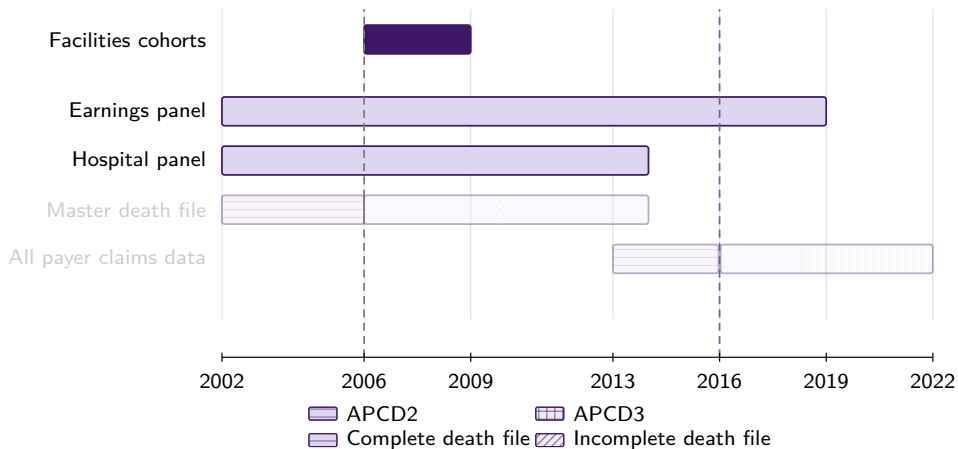
To assess the extent of this bias, we examine the “spike” in hospitalization in the quarter immediately preceding separation.

Is the spike an anticipation effect? We do not think so: there is no spike among workers not displaced during the mass layoff.

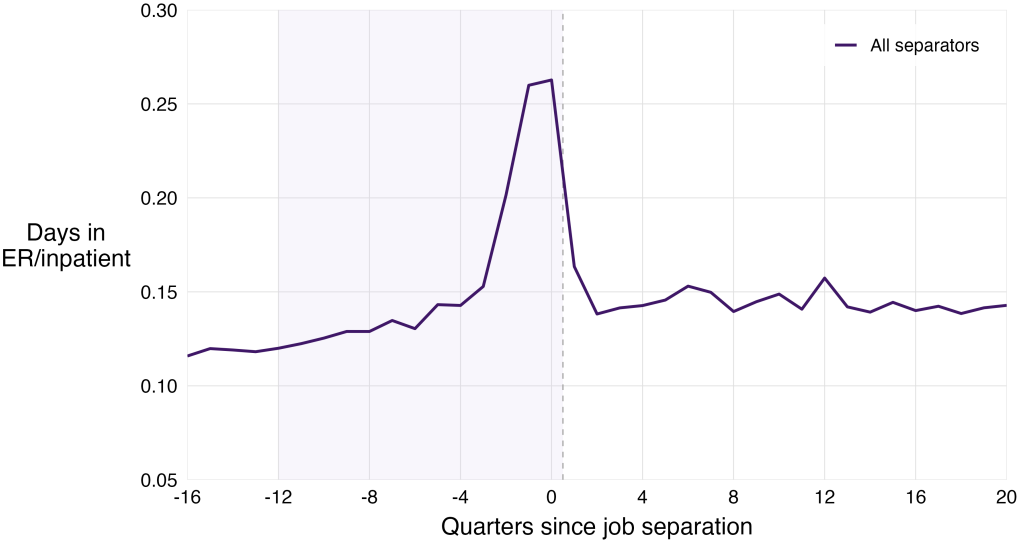
Two solutions

1. Match on measures of health before job loss
2. Use an intent-to-treat design

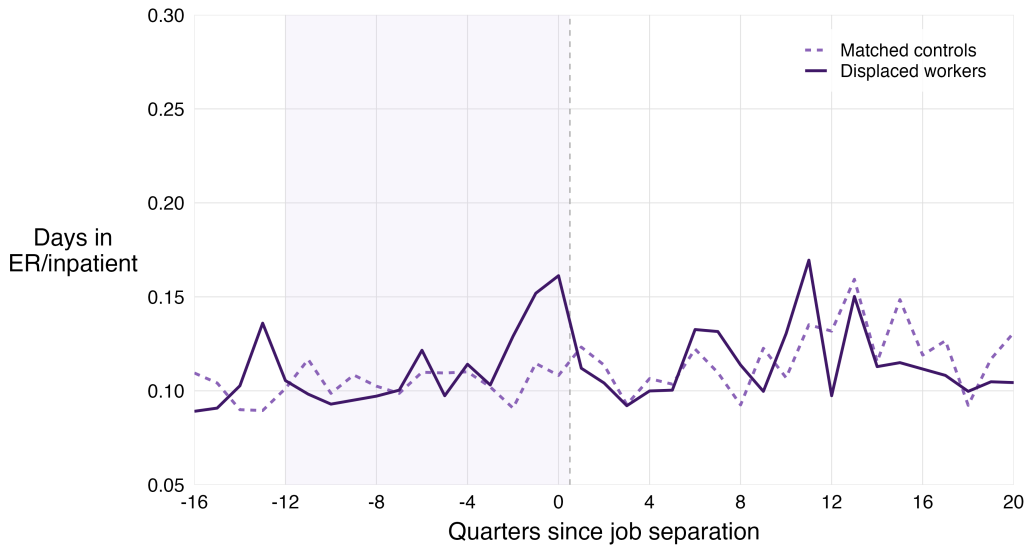
Data coverage: reverse-causality analysis



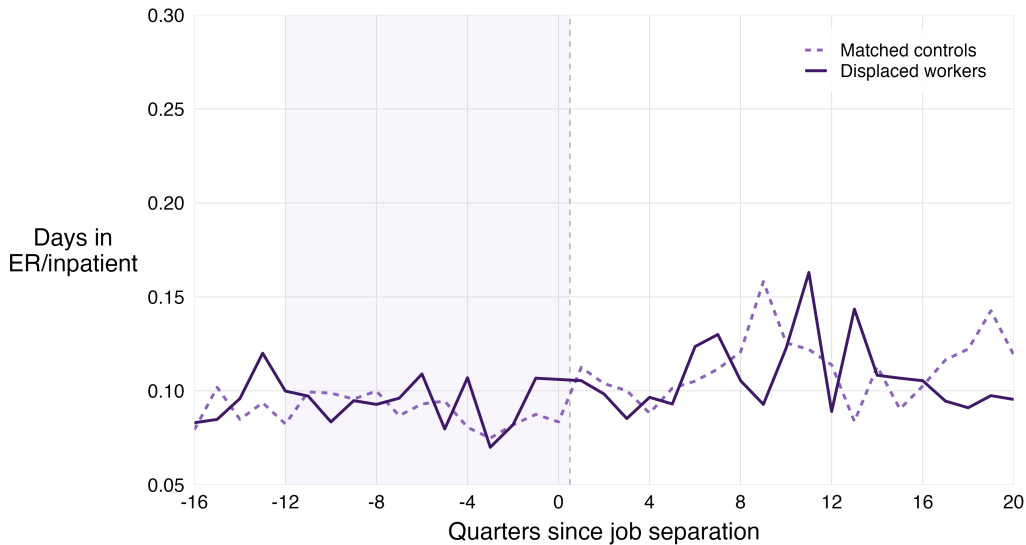
All separations



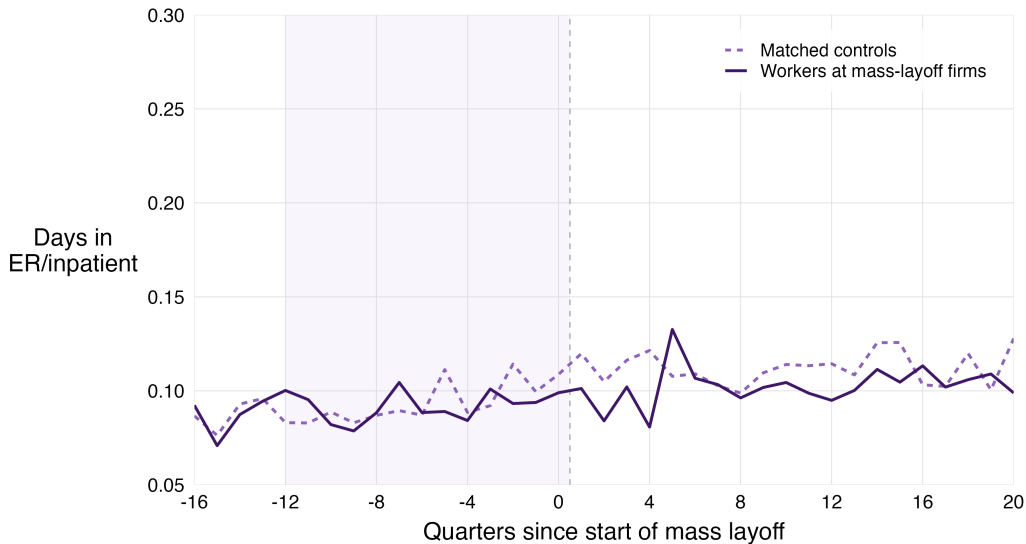
Mass-layoff separations



Mass layoffs matched on prior health



Incumbent workers at mass layoff firms



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Firms may do so to reduce costs, or because sick workers are less productive

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We estimate the following LPM on a sample including workers employed at mass layoff firms 2013–2015 prior to mass layoff and matched controls

$$Separate_i = \beta_1 MassLayoff_{j(i)} + \beta_2 Sick_i + \beta_3 Sick_i \times MassLayoff_{j(i)} + X_i\delta + \varepsilon_i$$

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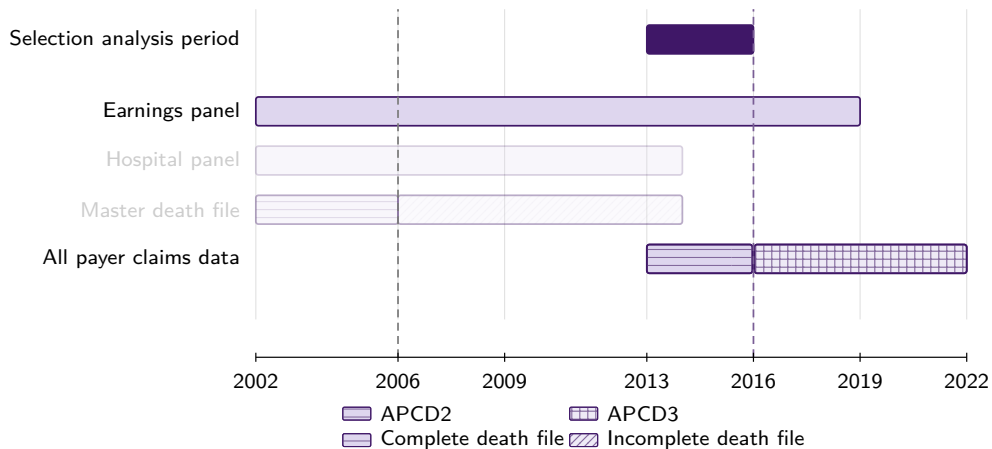
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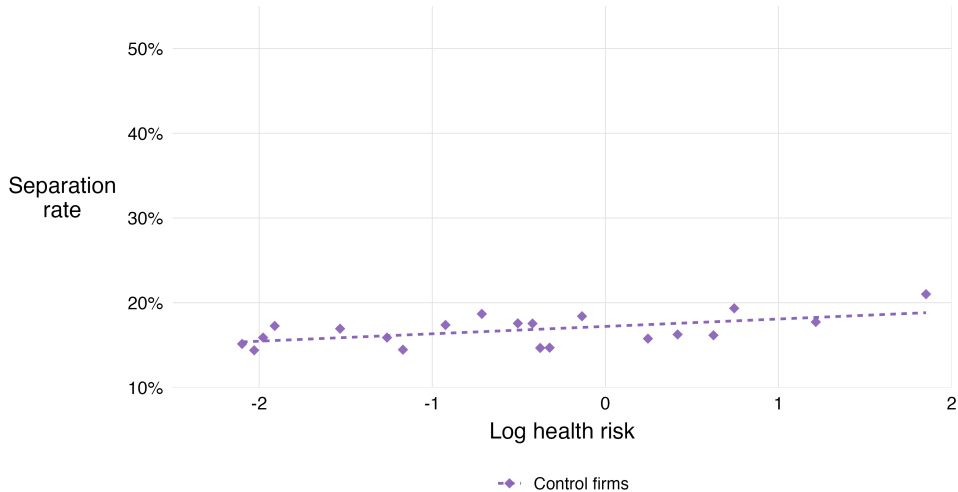
- ▶ $Sick_i$ measured by log of ACG predicted risk score: robust to other measures
- ▶ β_2 is the health-separation gradient in absence of mass layoffs
- ▶ β_3 captures the difference in this gradient during mass layoffs
 - ▶ Are the marginal workers laid off during a mass layoff sicker, on average?

Data coverage: selective-layoff analysis



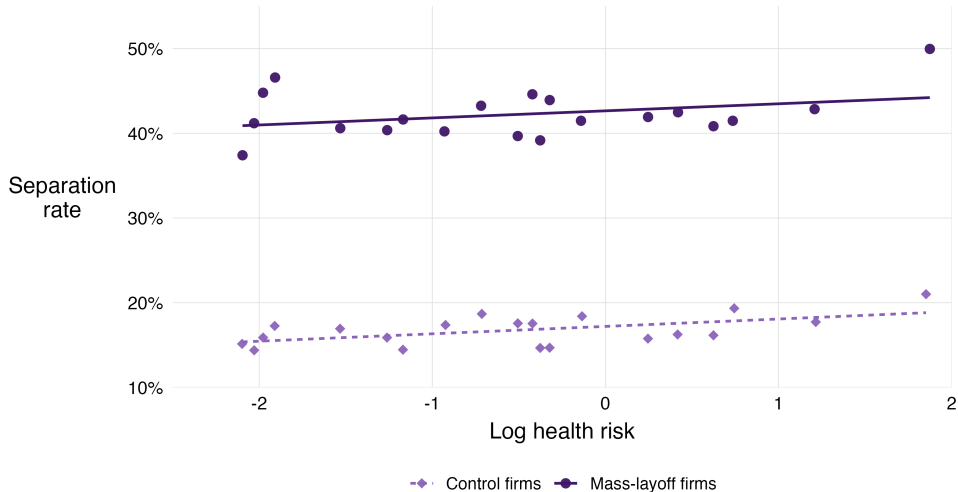
Do firms lay off sick workers during mass layoffs?

Separation-rate slope: control firms = +1.3 pp per log-point higher health risk



Do firms lay off sick workers during mass layoffs?

Separation-rate slope: control firms = +1.3 pp; mass-layoff firms = +0.9 pp per log point



Do firms lay off sick workers during mass layoffs?

	(1)	(2)	(3)	(4)	(5)	(6)
Mass-layoff firm	0.254 (0.005)	0.255 (0.005)	0.256 (0.005)	0.252 (0.005)	0.252 (0.005)	0.252 (0.005)
Log health risk	0.009 (0.002)	0.013 (0.002)	0.014 (0.002)	0.014 (0.002)	0.013 (0.002)	0.013 (0.002)
Mass-layoff firm $\times$ log health risk	0.000 (0.004)	-0.000 (0.004)	-0.002 (0.004)	-0.004 (0.004)	-0.004 (0.004)	-0.004 (0.004)
Observations	45,197	45,197	45,197	45,197	45,197	45,197
Age quintic, male		Y	Y	Y	Y	Y
Tenure FE, experience quintic			Y	Y	Y	Y
Log earnings, log firm size				Y	Y	Y
Cohort FE					Y	Y
Dependents FE						Y

Mass layoff regressions

Event study

$$y_{it} = \sum_{k \neq -8} \delta_k \mathbf{1}\{t - q(i) = k\} + \sum_{k \neq -8} \tau_k \mathbf{1}\{t - q(i) = k\} \times Displaced_i + \alpha_i + \gamma_t + X_{it}\beta + \varepsilon_{it}$$

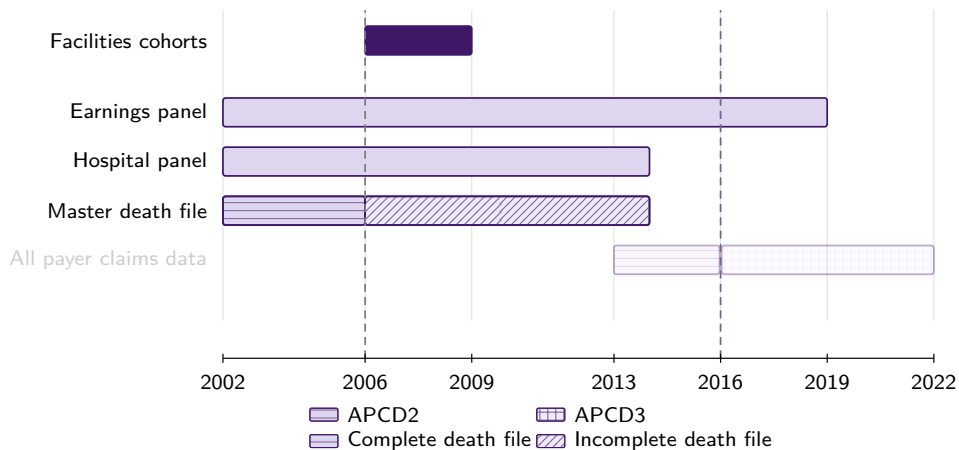
- ▶ Normalize $\sum_{k=-12}^{-4} \tau_k = 0$
- ▶ X_{it} is a quartic in age
- ▶ Standard errors clustered by worker

Difference-in-differences

$$\bar{y}_{ip} = \alpha_i + \alpha_p + \tau(Displaced_i \times Post_p) + \bar{X}_{ip}\beta + u_{ip}$$

- ▶ $p \in \{\text{pre: quarters 20–14 before job loss, post: quarters 1–20 after job loss}\}$
- ▶ $\bar{y}_{ip}$ is the mean of y_{it} in period p

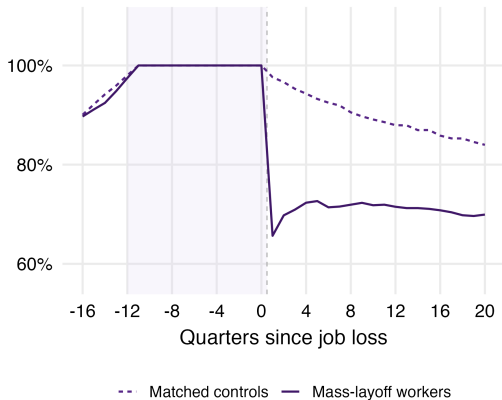
Data coverage: analysis of labor market, mortality, and hospital data



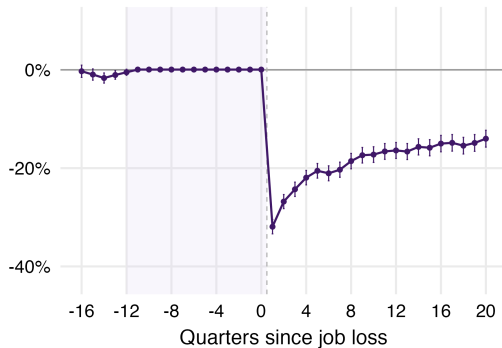
Job loss leads to persistent declines in employment

Dependent variable = 1 if employed in a given quarter

Raw means

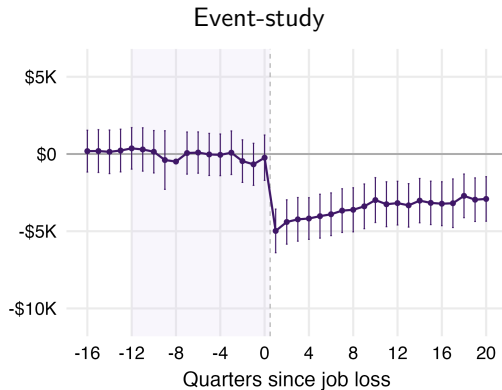
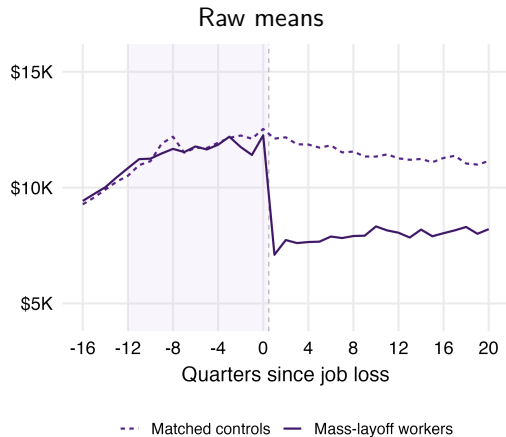


Event-study



Job loss leads to persistent declines in earnings

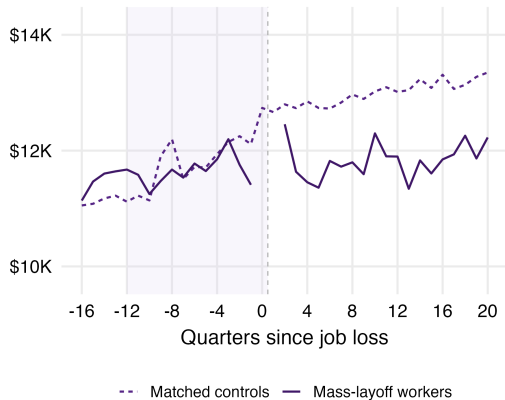
Dependent variable = total labor earnings per quarter



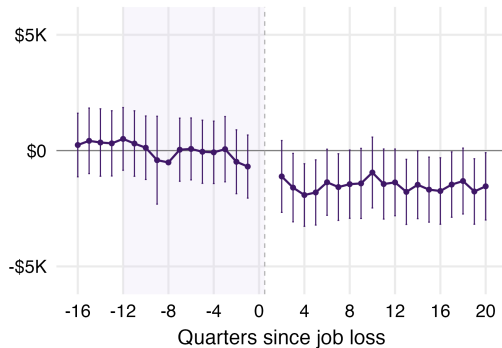
Job loss leads to persistent declines in wages

Dependent variable = earnings in “sandwich quarters”

Raw means



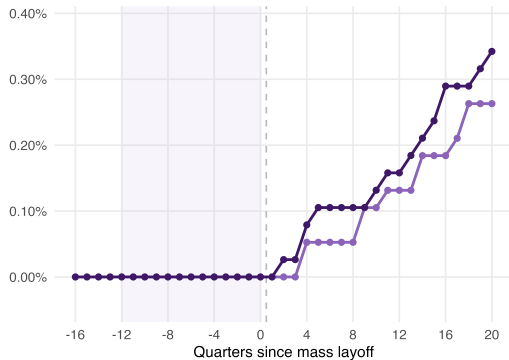
Event-study



Job loss has no detectable effect on mortality

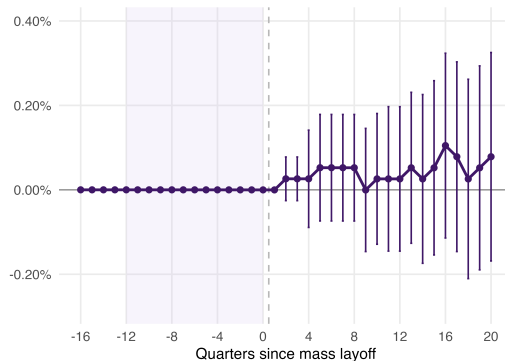
Dependent variable = 1 if dead

Raw means



—○— Matched controls —●— Mass-layoff workers

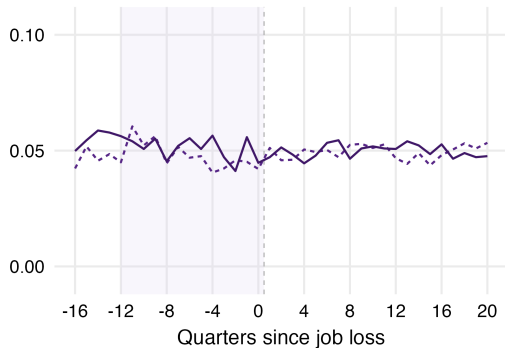
Event-study



Job loss has no effect on ER/inpatient hospitalization

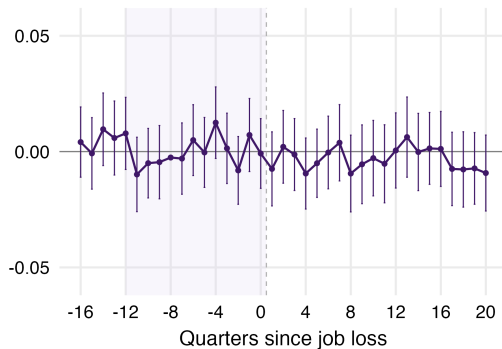
Dependent variable = ER/inpatient visits per quarter

Raw means



-- Matched controls — Displaced workers

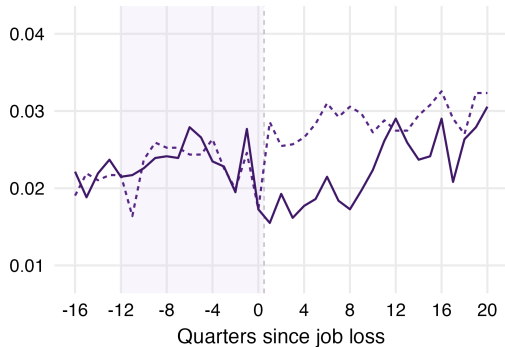
Event-study



Job loss decreases ambulatory hospital visits

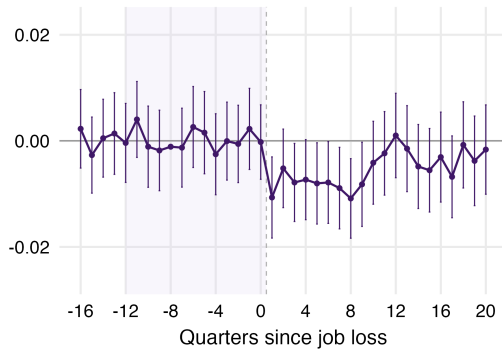
Dependent variable = ambulatory visits per quarter

Raw means



-- Matched controls — Displaced workers

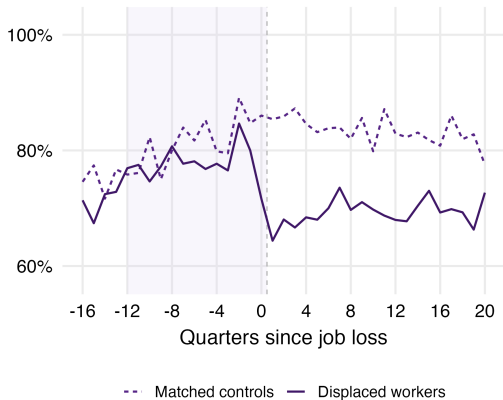
Event-study



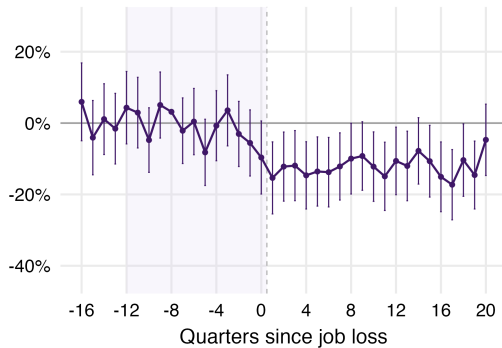
Hospital visits less likely to be covered by private insurance

Dependent variable = share of hospital visits listing private insurance as primary payer

Raw means



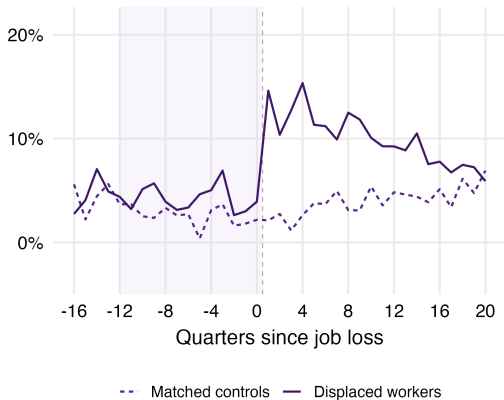
Event-study



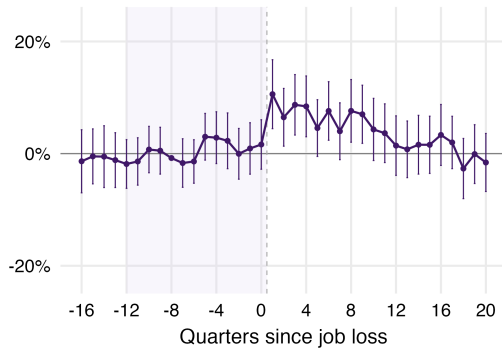
More paying-out-of-pocket

Dependent variable = share of hospital visits listing self-pay as primary payer

Raw means



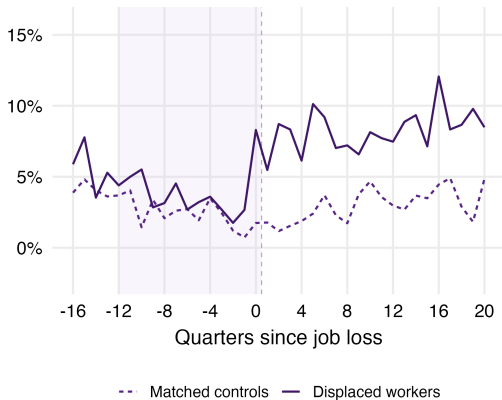
Event-study



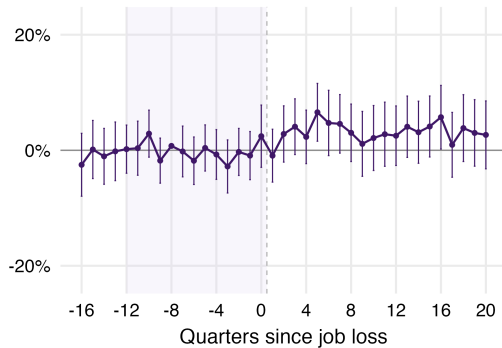
More Medicaid

Dependent variable = share of hospital visits listing Medicaid as primary payer

Raw means

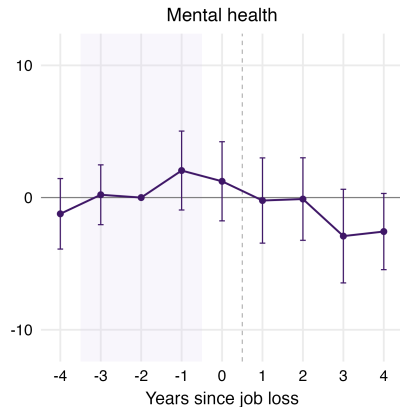
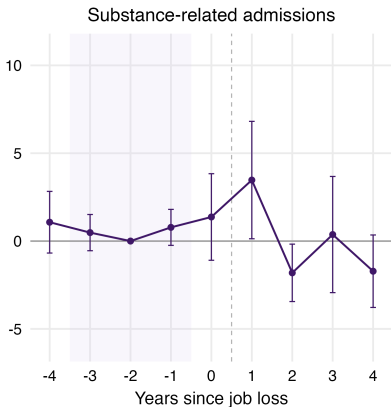


Event-study



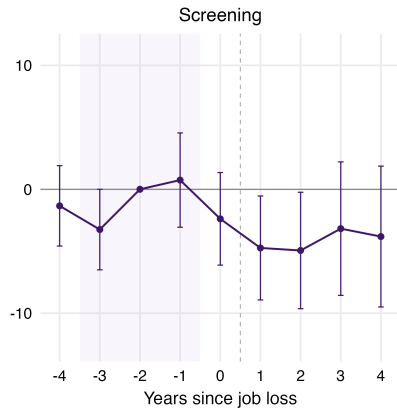
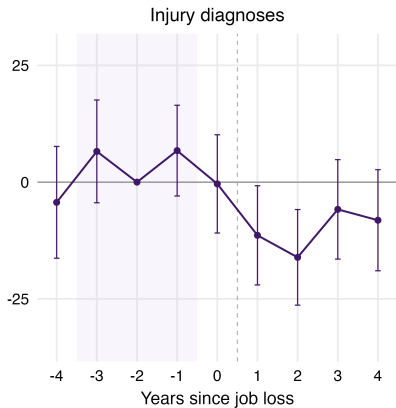
Effect on substance and mental health related hospitalization

Dependent variable = visits per 1,000 worker-quarters



Effect on hospital visits for injuries or screening

Dependent variable = visits per 1,000 worker-quarters



Recap

After job loss. . .

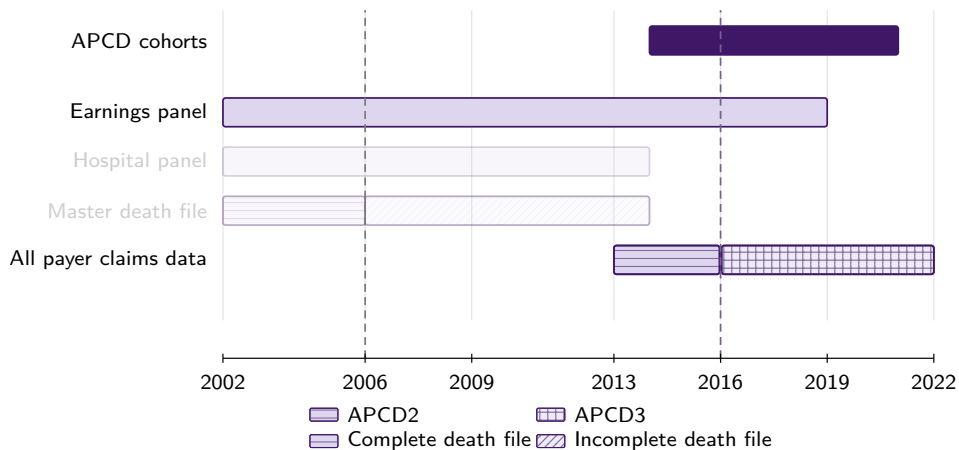
- ▶ Employment falls by 19 percentage points
- ▶ Quarterly earnings fall by about \$3,500 (baseline mean: \$11,500)
- ▶ No effect on mortality
- ▶ No effect on ER or inpatient hospitalization: -0.003 per quarter (baseline mean: 0.050)
 - ▶ One exception: substance-related visits rise by 3.5 per 1,000 in the short run (baseline mean: 0.5)
- ▶ Ambulatory hospital visits fall by 0.0068 per quarter (baseline mean: 0.0237)
 - ▶ Driven by reduced consumption of deferrable care
 - ▶ Screening visits fall by 4.7 (baseline mean: 4.9)
- ▶ Who pays at the hospital?
 - ▶ Private payer: -12 percentage points
 - ▶ Self-pay: $+4$ percentage points; Medicaid: $+3$ percentage points

Our interpretation: workers cut back on some deferrable forms of health care after job loss. With the exception of a short-run increase in hospitalization for substance abuse, there is no increase in acute negative health shocks.

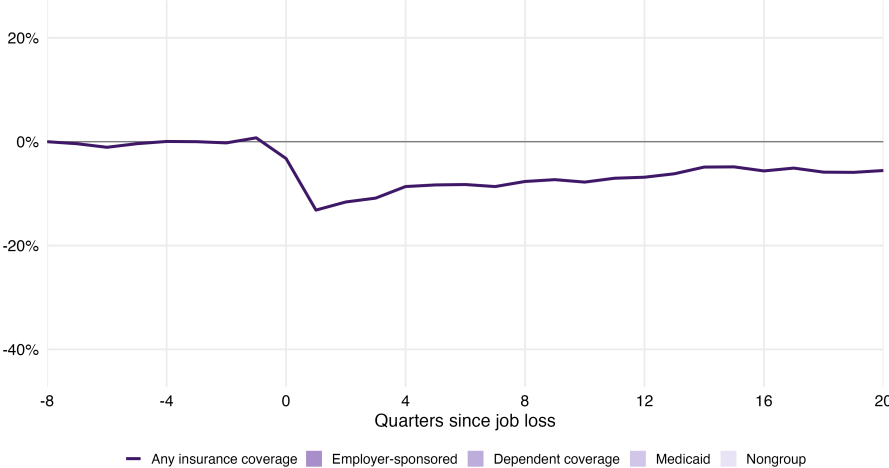
Analysis of claims data

- ▶ Up until this point, we have only looked at the effect of job loss on variables derived from mortality and hospital data
- ▶ Now, we start to analyze variables constructed from insurance claims
- ▶ We begin by estimating the effect of job loss on insurance
- ▶ Then, we estimate the effect of job loss on consumption of health care services
- ▶ In the final section, we try to use claims data to infer the effect of job loss on health
 - ▶ This is challenging because of selection bias: we only observe claims for insured individuals!
 - ▶ We use three different research designs to get around it

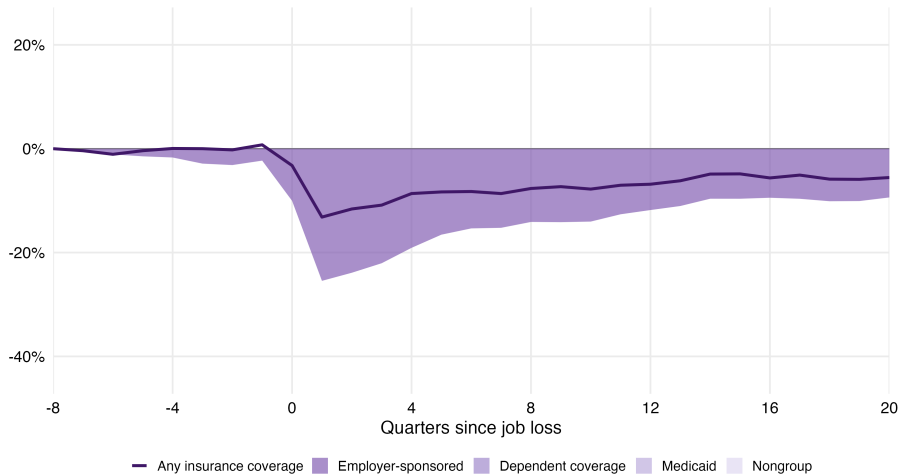
Data coverage: analysis of claims data



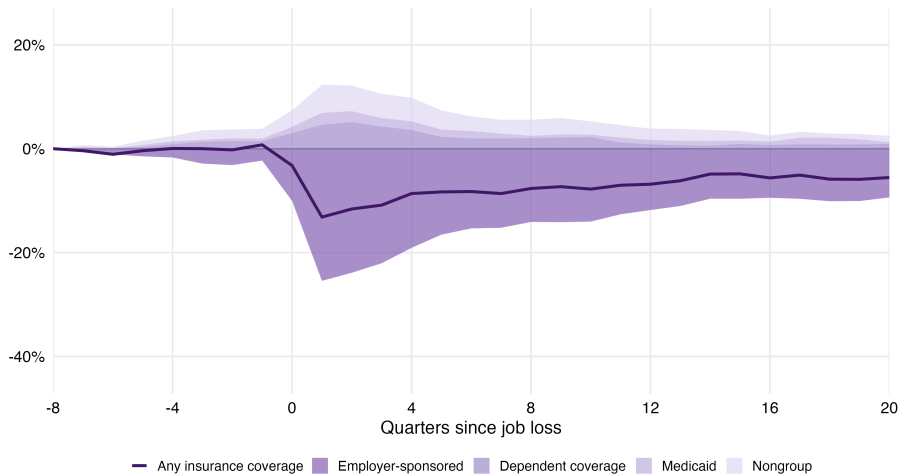
Job loss leads to persistent losses of health insurance coverage



Job loss leads to persistent losses of health insurance coverage

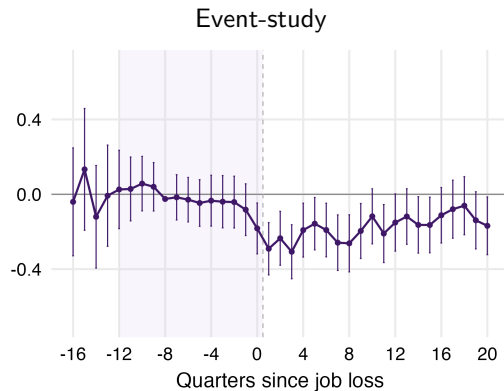
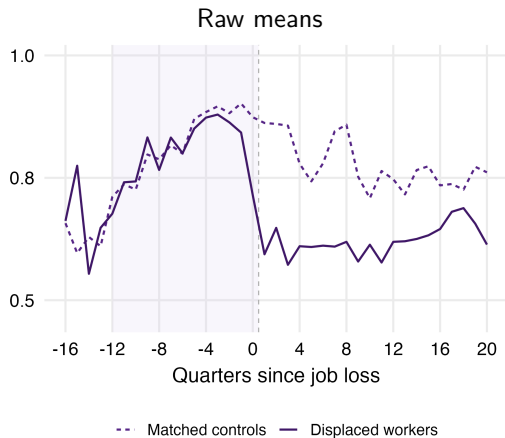


Job loss leads to persistent losses of health insurance coverage



Job loss decreases number of visits to the doctor

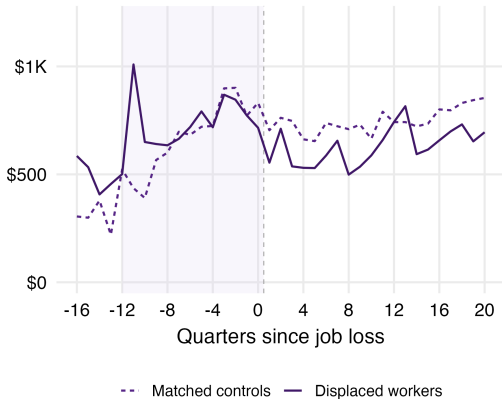
Dependent variable = professional-service visits per quarter



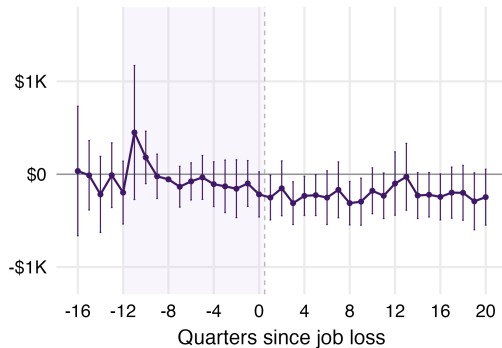
Job loss decreases total health care spending

Dependent variable = total allowed spending per quarter

Raw means



Event-study



Effect of job loss on types of care received by diagnosis group

Outcome	Pre-period control mean	Diff-in-diff levels	Diff-in-diff % Chg
Musculoskeletal	21.1	-4.1***	-19.4***
Skin	15.4	-1.6*	-10.2*
General surgery	14.4	-1.4*	-9.8*
Respiratory	14.3	-3.0***	-21.3***
Psychosocial	13.2	-2.7***	-20.2***
Ear, nose, throat	13.2	-2.0**	-15.2**
Cardiovascular	13.1	-2.2***	-16.6***
General signs and symptoms	11.5	-2.7***	-23.3***
Neurologic	11.3	-2.8***	-24.7***
Eye	11.1	-0.7	-6.1
Gastrointestinal/hepatic	9.1	-2.4***	-26.7***
Endocrine	8.6	-1.1*	-12.4*
Genito-urinary	7.6	-0.2	-2.2
Allergy	7.4	-0.6	-7.8
Female reproductive	7.4	-0.7	-9.0
Nutrition	5.0	-2.3***	-45.9***

Effect of job loss on types of care received

Outcome	Pre-period control mean	Diff-in-diff levels	Diff-in-diff % Chg
Discretionary / prevention / admin	38.2	-6.3***	-16.5***
Signs / symptoms	36.1	-5.1***	-14.2***
Chronic or progressive	29.6	-4.0***	-13.5***
Time limited	28.1	-4.8***	-17.2***
Likely to recur	20.6	-3.0***	-14.4***
Psychosocial	13.4	-2.7***	-20.2***
Injuries/adverse effects: minor	6.5	-1.4**	-21.7**
Injuries/adverse effects: major	5.3	-0.9*	-16.3*
Pregnancy	2.3	-0.1	-5.0

Inferring effect of job loss on health from insurance claims

Econometrically challenging due to a selection problem: we only observe claims for people who are insured!

We try three different research designs

Design #1: wait a few years after job loss and then condition on being insured

- ▶ Selecting on a post-treatment outcome, but for both displaced and control workers

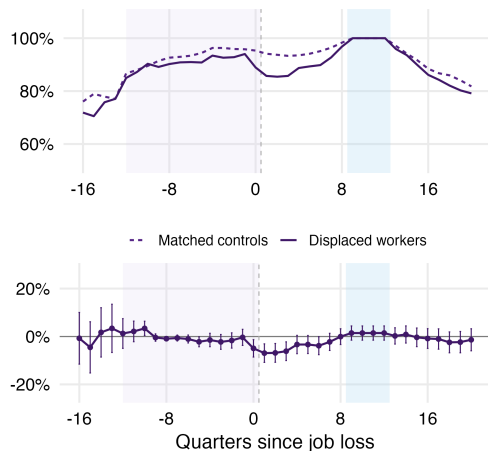
Design #2: triple difference using workers insured through dependents

- ▶ Job loss has less effect on insurance for those with insurance through a spouse or family member

Design #3: compare outcomes among individuals age 65+ on Medicare Advantage

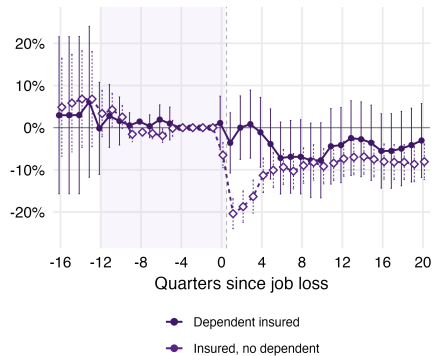
- ▶ The only form of Medicare in our data
- ▶ We observe workers laid off in their late forties and fifties
- ▶ Compare with matched control workers who were not laid off but also ended up on Medicare Advantage plans

Design #1: wait until reinsured



Outcome	Pre-period control mean	Q9–Q12 Diff-in-diff
Chronic conditions	0.865	-0.011 (0.083)
ACG risk	0.842	0.049 (0.100)
Annual total spending	3,818	-539 (807)
ACG pharmacy cost	602	-18 (280)
ER claims	0.022	0.010 (0.007)
Inpatient claims	0.007	0.001 (0.004)
Psych condition	0.178	0.012 (0.019)
Substance condition	0.009	0.003 (0.005)
Workers		1,996
Observations		3,834

Design #2: Dependent-insurance



Outcome	Pre-period mean		Diff-in-diff		Triple difference
	Dep.	No dep.	Dep.	No dep.	Dep. — No dep.
Insured	0.982	0.986	-0.036 (0.027)	-0.130*** (0.013)	0.094*** (0.030)
Total spending	706	890	153 (146)	-211* (111)	363** (184)
Chronic conditions	0.871	0.892	-0.096 (0.107)	-0.114** (0.051)	0.018 (0.118)
Health risk	0.780	0.876	-0.118 (0.145)	-0.068 (0.058)	-0.050 (0.156)

Design #3: Medicare Advantage

Outcome	t-2 (%)		t-1 (%)	
	Displaced	Control	Displaced	Control
Any ER/inpatient use	4.0	3.8	3.8	3.6
Any ambulatory use	5.1	5.0	5.2	5.1

Outcome	Mean		Displaced – control		
	Displaced	Control	Unadjusted	Baseline	+ health
Chronic conditions	2.813	2.674	0.138 (0.153)	0.208 (0.157)	0.142 (0.146)
ACG risk	2.191	2.060	0.131 (0.134)	0.170 (0.136)	0.122 (0.128)
Total spending	1,522	1,545	-23 (152)	-37 (158)	-90 (152)
Pharmacy cost	1,784	1,749	34 (225)	-23 (243)	-130 (233)
ER claims	0.048	0.039	0.010 (0.007)	0.009 (0.007)	0.009 (0.007)
Inpatient claims	0.015	0.015	0.000 (0.003)	0.000 (0.003)	-0.000 (0.003)
Psych condition	0.299	0.289	0.010 (0.021)	0.023 (0.021)	0.020 (0.021)
Substance condition	0.030	0.019	0.012 (0.006)	0.012 (0.007)	0.009 (0.007)

Conclusion

Our evidence is mostly consistent with RCTs and with the literature on recessions and health, but inconsistent with previous US evidence from early 1980s Pennsylvania.

Once reverse causality bias is taken care of, there is little evidence of broad effects of job loss on measured health.

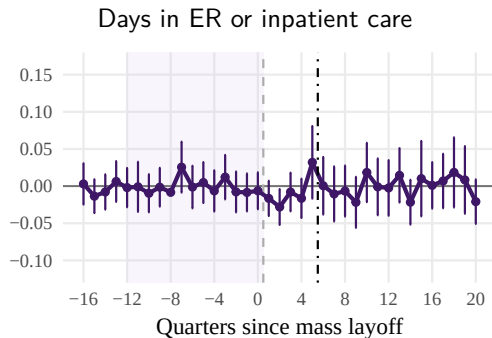
Resolution to the health puzzle I started with: there is no puzzle?

Merci!

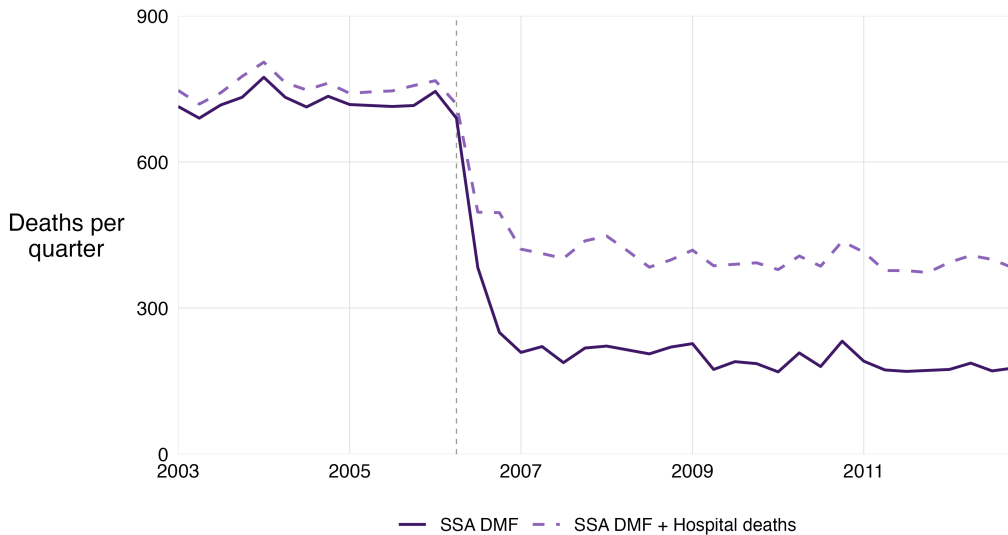
Appendix

Appendix

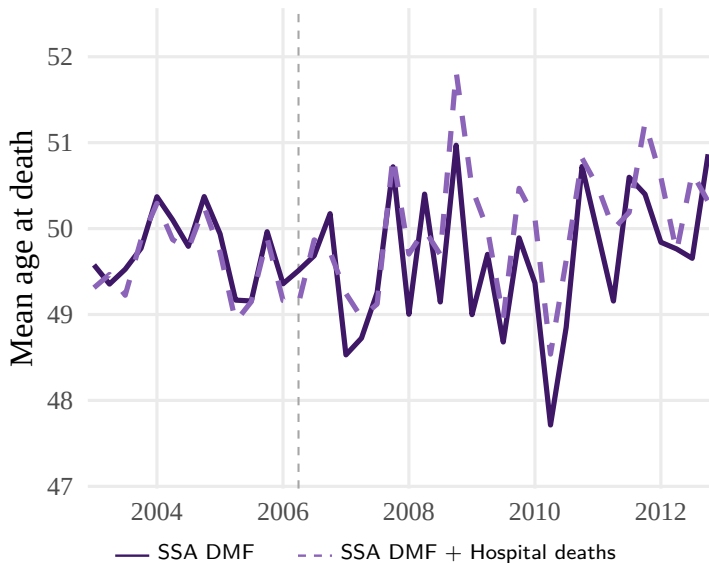
No pre-layoff spike among non-displaced workers



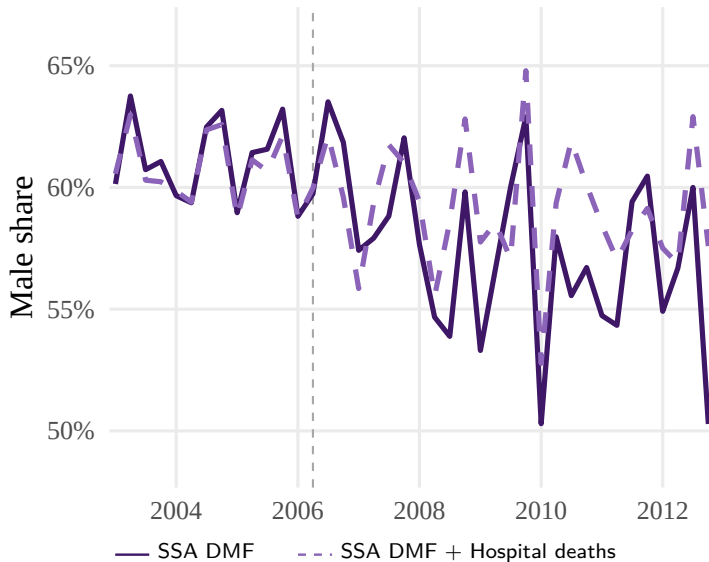
Mortality data quality: deaths per quarter



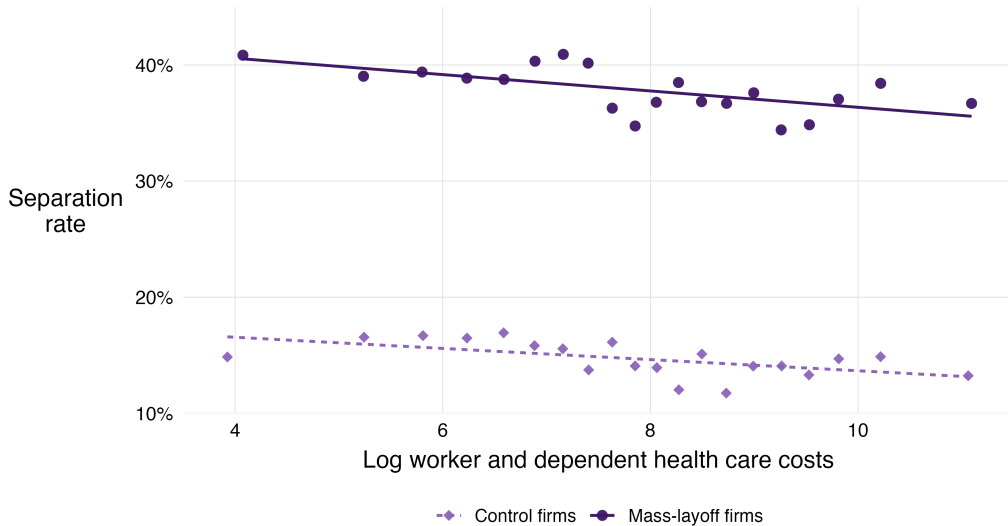
Mortality data quality: mean age at death



Mortality data quality: male share of decedents



Do firms lay off costly workers in mass layoffs?



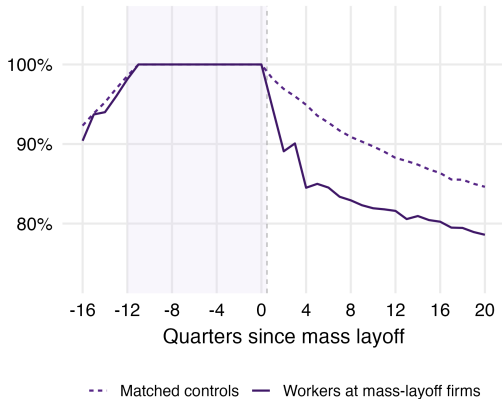
Do firms lay off costly workers in mass layoffs?

	(1)	(2)	(3)	(4)	(5)	(6)
Mass-layoff firm	0.252 (0.021)	0.255 (0.021)	0.253 (0.021)	0.255 (0.021)	0.256 (0.021)	0.254 (0.021)
Log worker and dependent health care costs	-0.004 (0.001)	-0.001 (0.002)	0.000 (0.001)	0.002 (0.002)	0.002 (0.002)	0.006 (0.002)
Mass-layoff firm $\times$ log costs	-0.003 (0.003)	-0.003 (0.003)	-0.002 (0.003)	-0.002 (0.003)	-0.003 (0.003)	-0.002 (0.003)
Observations	36,408	36,408	36,408	36,408	36,408	36,408
Age quintic, male		Y	Y	Y	Y	Y
Tenure FE, experience quintic			Y	Y	Y	Y
Log earnings, log firm size				Y	Y	Y
Cohort FE					Y	Y
Dependents FE						Y

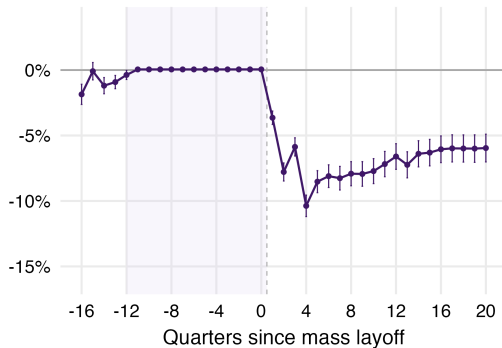
Job loss leads to persistent declines in employment (intent-to-treat)

Dependent variable = 1 if employed in a given quarter

Raw means

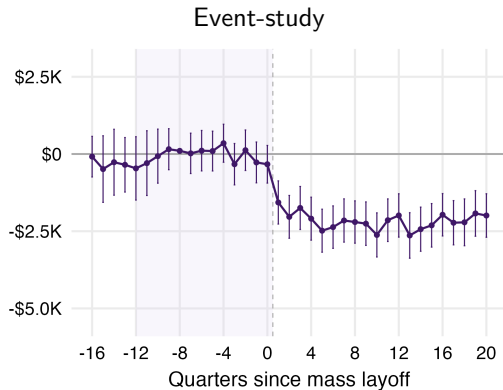
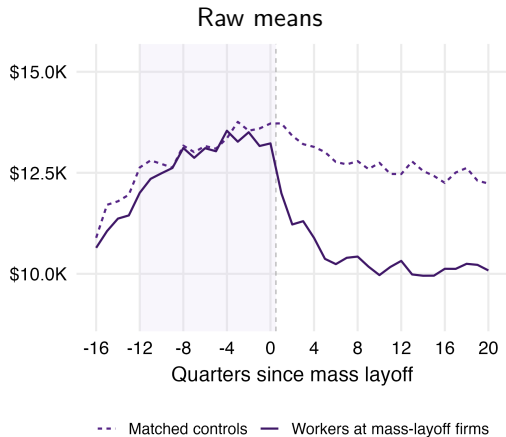


Event-study



Job loss leads to persistent declines in earnings (intent-to-treat)

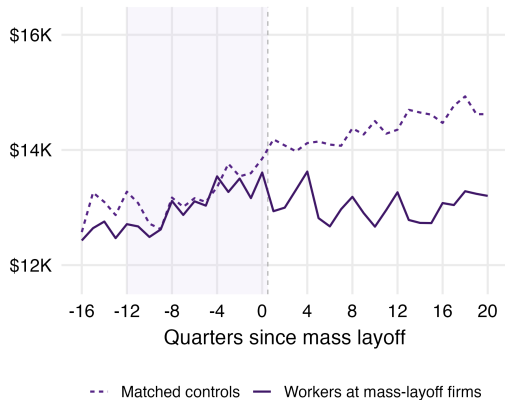
Dependent variable = total labor earnings per quarter



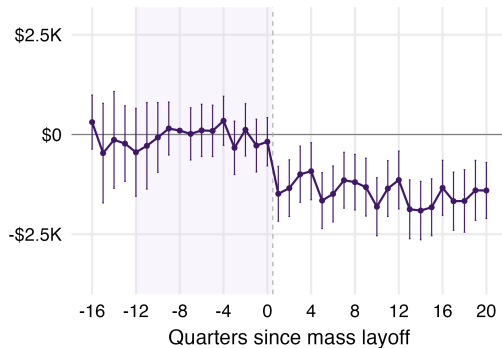
Job loss leads to persistent declines in wages (intent-to-treat)

Dependent variable = earnings in “sandwich quarters”

Raw means

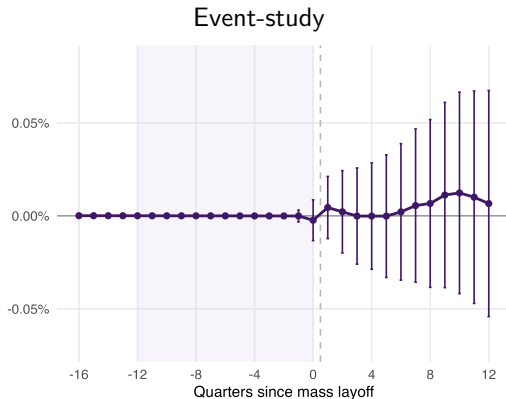
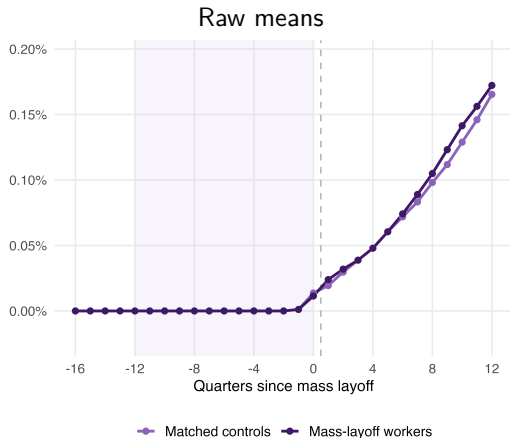


Event-study



Job loss has no detectable effect on mortality (intent-to-treat)

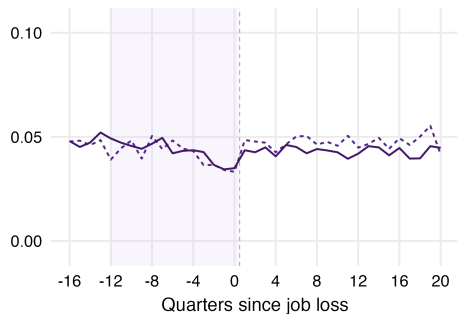
Dependent variable = 1 if dead



Job loss has no effect on ER/inpatient hospitalization

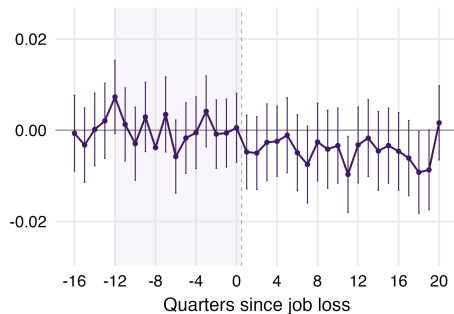
Intent-to-treat; dependent variable = ER/inpatient visits per quarter

Raw means



-- Matched controls — Workers at mass-layoff firms

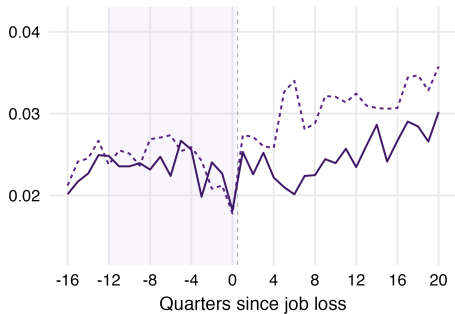
Event-study



Job loss decreases ambulatory hospital visits

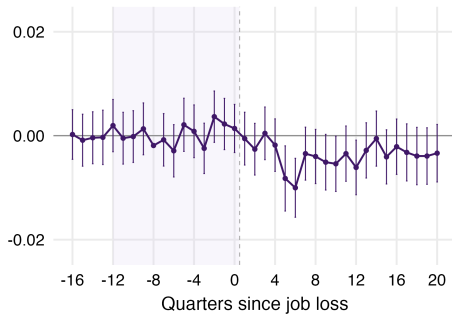
Intent-to-treat; dependent variable = ambulatory visits per quarter

Raw means



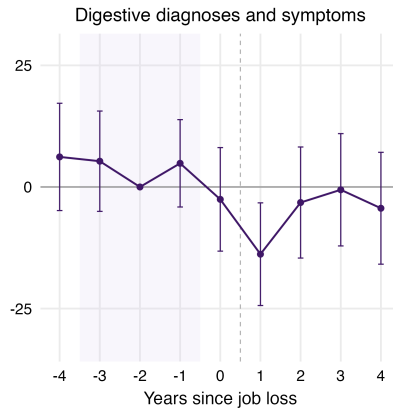
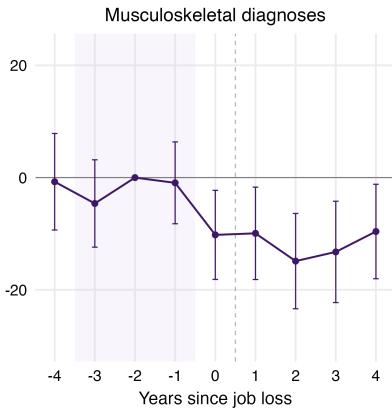
-- Matched controls — Workers at mass-layoff firms

Event-study



Job loss increases visits for specific diagnoses

Dependent variable = visits per 1,000 worker-quarters



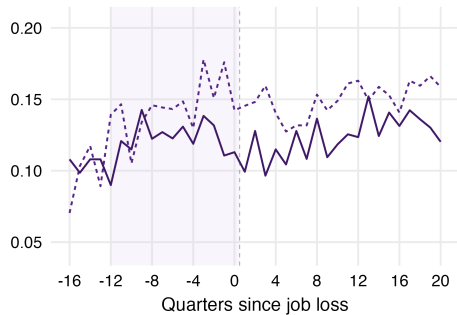
Effect of job loss on hospitalization by diagnosis

	Visits per 1,000 worker-quarters			Share of hospital visits	
	Pre-period control mean	Diff-in-diff	Diff-in-diff %	Share	Diff
Total visits	73.1	-9.9***	-13.5***	100.0	0.0
Injury	14.5	-2.5**	-17.6**	19.8	-1.2
Digestive / abdominal	13.2	-2.1	-15.8	18.0	-0.5
Musculoskeletal	7.6	-0.8	-11.1	10.4	0.2
Pregnancy, delivery, newborn	6.9	-0.6	-8.6	9.5	0.4
Respiratory / chest	3.9	-0.5	-11.5	5.4	0.2
Female reproductive conditions	3.9	-0.1	-1.9	5.3	0.5
Eye and ENT	3.8	-0.4	-9.5	5.2	0.1
Genitourinary	2.4	-0.2	-6.7	3.3	0.3
Cardiovascular / circulatory	2.1	-0.9	-43.5	2.8	-0.8
Skin	1.8	0.3	14.2	2.5	0.7
Screening	1.3	-1.0***	-76.6***	1.8	-0.9*
Workers	8,964			6,056	
Observations	18,064			8,283	

Job loss decreases outpatient visits

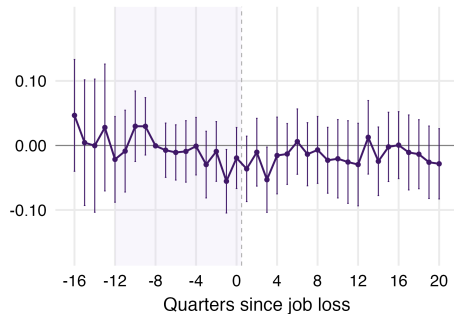
Dependent variable = outpatient visits per quarter

Raw means



-- Matched controls — Displaced workers

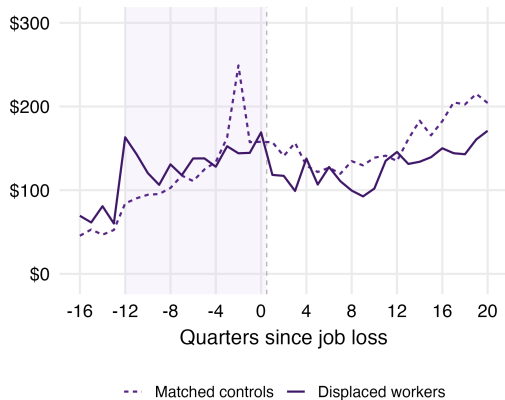
Event-study



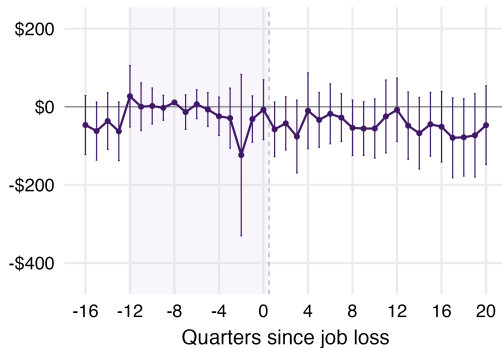
Job loss decreases pharmacy spending

Dependent variable = pharmacy spending per quarter

Raw means



Event-study



Difference-in-differences estimates in post-loss insured samples

Outcome	Pre-period control mean	Q5–Q8	Q9–Q12	Q13–Q16	Y2–Y5	Y3–Y5	Y4–Y5
Chronic conditions	0.865	0.063 (0.068)	-0.011 (0.083)	-0.024 (0.090)	-0.101 (0.082)	-0.025 (0.086)	-0.042 (0.080)
ACG risk	0.842	0.104 (0.091)	0.049 (0.100)	-0.050 (0.104)	-0.049 (0.095)	-0.020 (0.095)	-0.049 (0.089)
Annual total spending	3,818	133 (562)	-539 (807)	-592 (744)	-1,202 (820)	-730 (732)	-452 (694)
ACG pharmacy cost	602	28 (205)	-18 (280)	-592 (475)	-165 (481)	-582 (466)	-363 (414)
ER claims	0.022	0.010* (0.006)	0.010 (0.007)	0.014* (0.007)	0.010 (0.006)	0.007 (0.008)	0.009 (0.006)
Inpatient claims	0.007	0.005 (0.003)	0.001 (0.004)	0.005 (0.006)	0.001 (0.004)	0.002 (0.004)	0.002 (0.004)
Psych condition	0.178	0.018 (0.016)	0.012 (0.019)	0.025 (0.019)	0.005 (0.018)	0.007 (0.018)	0.003 (0.018)
Substance condition	0.009	0.008** (0.004)	0.003 (0.005)	-0.001 (0.005)	0.007 (0.005)	-0.001 (0.005)	-0.003 (0.005)
Workers Observations		2,151 4,078	1,996 3,834	1,987 3,824	1,666 3,178	1,801 3,440	2,039 3,902